Software as a Service

- **SaaS (Software as a Service)** describes any **cloud service** where **consumers** are able to **access software applications** over the **internet**
  - **Google**, **Twitter** and **Facebook** are all **examples of SaaS**, with users able to access the services via any internet enabled device
  - **Enterprise users** are able to use applications for a range of needs, including **accounting** and **invoicing**, **tracking sales**, **planning**, **performance monitoring** and **communications**
- SaaS is often referred to as **software-on-demand** and utilizing it as **renting software rather than buying it**
- With traditional software applications you would purchase the software upfront as a package and then install it onto your computer

Software as a Service cont.

- The **software's license** may also **limit the number of users** and/or **devices where the software can be deployed**
- **Software as a Service users**, however, **subscribe** to the software rather than purchase it**, usually on a monthly basis**
- **Applications are purchased** and **used online** with files (**data files**) **saved in the cloud** rather than on individual computers

Software as a Service Cont.

Benefits of SaaS to organizations and personal users

No additional hardware costs	Updates are automated
No initial setup costs	Accessible from any location
Pay for what you use	Cross device compatibility
Usage is scalable	Applications can be customized

Software as a Service Cont.

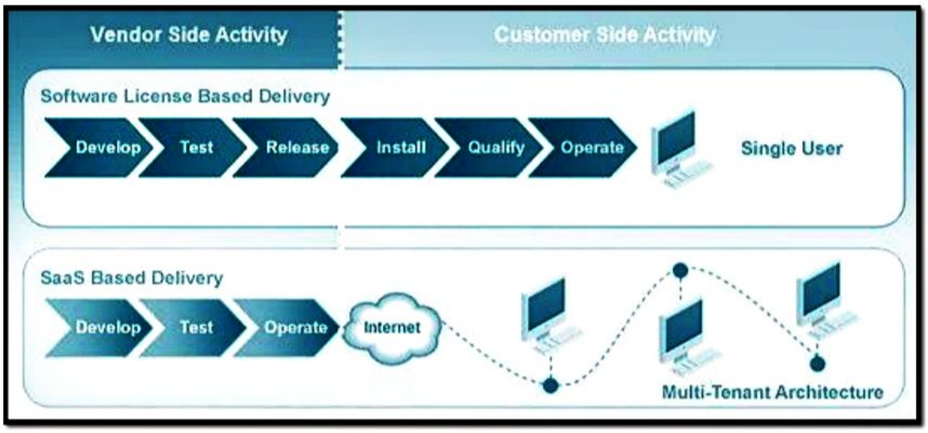
- **Office software** is the best **example** of **businesses utilizing SaaS**
- Tasks related to accounting, invoicing, sales and planning can all be performed through Software as a Service.
- The required software can be subscribed via the internet and then accessed online via any computer in the office using a username and password.
- If needs change they can easily switch to software that better meets their requirements.
- Everyone who needs access to a particular piece of software can be set up as a user
- Whether it is one or two people or every employee in a corporation.

SaaS Summary

- There **are no setup costs** with SaaS, as there often are with other applications
- SaaS **is scalable with upgrades** available **on demand**
- Access to Software as a Service is **compatible across all internet enabled devices**
- As long as there is an internet connection, **applications are accessible from any location**

SaaS Architecture

Traditional Software Delivery Model



SaaS Model

SaaS Architecture (Cont…)

- The diagram illustrates the key **differences** between the **traditional software delivery** model and a **SaaS-based** delivery model
  - In essence, **all users of SaaS**-based applications **run exactly the same code** with customizations and configurations stored as metadata parameters
- The **key difference** between **SaaS** and **traditional software** delivery is that **SaaS-based applications** are **fully hosted within some form of Data Centre environment**
  - All updates to the SaaS application are carried out within the Data Centre and therefore the **user** can always be assured that they **are using the most recent version of the application**
  - These **applications** are designed to work and take advantage of **operating within a web browser-based environment**

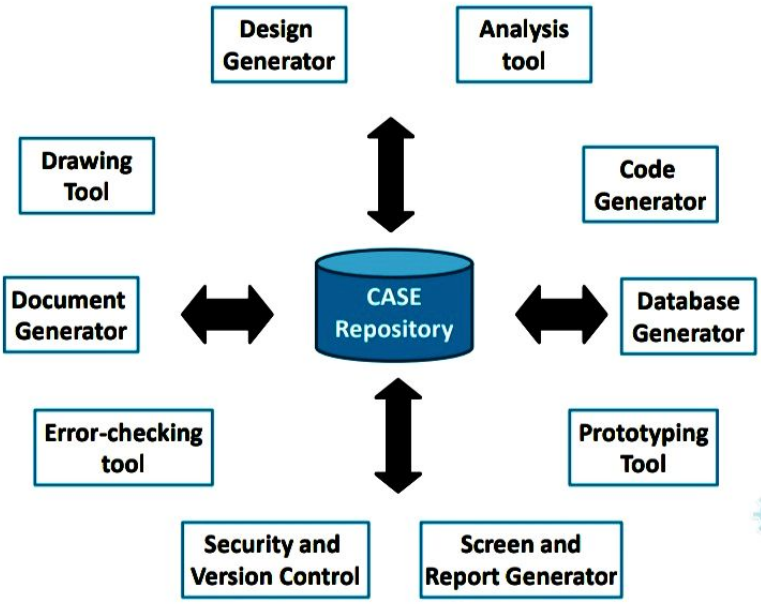
SaaS Architecture (Cont…)

- Since nearly every **computer comes with a browser** as part of the operating environment, there is **hardly any implementation work required by a company's IT department**
- This allows the IT department to focus on other activities **reducing support costs**
- **SaaS** is **traditionally sold** on a **subscription basis** that **includes** upgrades, maintenance and some level of basic support
- These subscriptions operate on a monthly, quarterly or yearly basis
- SaaS is **ideal for companies** who would be **using the application periodically**

Computer-Aided Software Engineering

- A **CASE (Computer Aided Software Engineering)** tool is a generic term used to denote **any form of automated support** for **software engineering**
- A **CASE tool** means any tool used to **automate some activity** associated with **software development**
- The primary **reasons** for using a **CASE tool** are:
  - To **increase productivity**
  - To **help produce better quality software** at lower cost

Components of CASE



Components of CASE Cont.

- **CASE repository**
  - **Central component** of any CASE tool
  - Also known as the **information repository** or **data dictionary**
  - **Centralized database**
  - Allows **easy sharing of information** between tools and SDLC activities
  - Used to **store graphical diagrams** and **prototype forms** and **reports** during analysis and design workflows
  - Provides **wealth of information** to **project manager** and allows control over project
  - Facilitates **reusability**

Components of CASE Cont.

- **Analysis tools**
  - Generate **reports** that help **identify possible inconsistencies**, **redundancies** and **omissions** (deficiency)
  - Generally focus on
    - **diagram completeness and consistency**
    - **data structures** and **usage**
- **CASE documentation generator tools**
  - **Create standard reports** based on **contents of repository**
  - Need **textual descriptions** of requirements, solutions, diagrams of data and processes, prototype forms and reports, program specifications and user documentation
  - **High-quality documentation leads** to **80% reduction in system maintenance effort** in comparison to average quality documentation

CASE Tool Classification

- **Business process engineering tools**
  - represent business data objects, their relationships and flow of the data objects between company business areas
- **Process modeling and management tools**
  - represent key elements of processes and provide links to other tools that provide support to defined process activities
- **Project planning tools**
  - used for cost and effort estimation and project scheduling
- **Risk analysis tools**
  - help project managers build risk tables by providing detailed guidance in the identification and analysis of risks

CASE Tool Classification Cont.

- **Requirements tracing tools**
  - provide systematic database-like approach to tracking requirement status beginning with specification
- **Metrics and management tools**
  - management oriented tools capture project specific metrics that provide an overall indication of productivity or quality
- **Documentation tools**
  - provide opportunities for improved productivity by reducing the amount of time needed to produce work products

CASE Tool Classification Cont.

- **System software tools**
  - network system software, object management services, distributed component support and communications software
- **Quality assurance tools**
  - metrics tools that audit source code to determine compliance with language standards or tools that extract metrics to project the quality of software
- **Database management tools**
  - RDBMS and OODMS (object-oriented database management system) serve as the foundation for the establishment of the CASE repository

CASE Tool Classification Cont.

- **Software configuration management tools**
  - uses the CASE repository to assist with all SCM tasks (identification, version control, change control, auditing, status accounting)
- **Analysis and design tools**
  - enable the software engineer to create analysis and design models of the system to be built, perform consistency checking between models
- **Simulation tools**
  - simulation tools provide software engineers with ability to predict the behavior of real-time systems before they are built and the creation of interface mockups for customer review

CASE Tool Classification Cont.

- **Interface design and development tools**
  - toolkits of interface components, often part environment with a GUI to allow rapid prototyping of user interface designs
- **Prototyping tools**
  - enable rapid definition of screen layouts, data design, and report generation
- **Programming tools**
  - compilers, editors, debuggers, OO programming environments, graphical programming environments, applications generators, and database query generators
- **Web development tools**
  - assist with the generation of web page text, graphics, forms, scripts, applets, etc.

CASE Tool Classification Cont.

- **Integration and testing tools**
  - data acquisition - get data for testing
  - static measurement - analyze source code without using test cases
  - dynamic measurement - analyze source code during execution
  - test management
- **Static analysis tools**
  - code-based testing tools, specialized testing languages, requirements-based testing tools
- **Dynamic analysis tools**
  - intrusive tools modify source code by inserting probes (enquiry) to check path coverage, assertions or execution flow

CASE Tool Classification Cont.

- **Test management tools**
  - coordinate regression testing, compare actual and expected output, conduct batch testing and serve as generic test drivers
- **Client/server testing tools**
  - exercise the GUI and network communications requirements for the client and server
- **Re-engineering tools**
  - reverse engineering to specification tools
    - generate analysis and design models from source code
  - code restructuring and analysis tools
    - analyze program syntax, generate control flow graph and automatically generates a structured program
  - on-line system reengineering tools
    - used to modify on-line DBMS

Summary

- Component-Based Software Engineering
- Client/Server Software Engineering
- Web Engineering
  - Framework for Web Engineering
- Computer-Aided Software Engineering
  - Components of CASE
  - CASE Tool Taxonomy